BANK CUSTOMER CHURN ANALYSIS

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1. Project Overview and Objective

The Bank Customer Churn Analysis project uses Power BI to analyze customer data and identify patterns that lead to customer attrition. The dataset includes key attributes such as credit score, geography, age, tenure, balance, and activity status, enabling a comprehensive understanding of customer behavior. Through data cleaning, modeling, and visualization, the project uncovers trends in churn across different segments and highlights high-risk customers. Interactive dashboards and DAX measures are used to monitor key performance indicators and support data-driven decision-making. Overall, the project helps businesses improve customer retention strategies, optimize services, and enhance overall customer satisfaction.

Project Objectives

- Analyze customer data to understand patterns and trends related to churn
- Identify key factors influencing customer attrition such as credit score, tenure, and activity status
- Segment customers based on demographics and behavior to detect high-risk groups
- Develop interactive Power BI dashboards to monitor key performance indicators (KPIs)
- Enable predictive insights to anticipate potential churn customers
- Provide data-driven recommendations to improve customer retention and business performance

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2. Data Sources

- **Source**: Free Sample Dataset Download - Bank Customer Churn - Maven Analytics | Build Data Skills, Faster
- **Timeline**: 2020–2025
- **Domain**: Banking / Customer Analytics

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3. Problem Statement

- To identify key factors influencing customer churn
- To analyze churn behavior based on age, gender, and balance
- To evaluate customer segments with high churn risk To help banks design retention strategies to reduce churn

4.Column Description:

Attribute Name	Data Type	Description
CustomerId	Integer	Unique ID for each customer
Surname	Text	Customer name
CreditScore	Integer	Customer credit score
Geography	Categorical	Country (France, Germany, Spain)
Gender	Categorical	Male/Female
Age	Integer	Customer age
Tenure	Integer	Years with bank
Balance	Decimal	Account balance
NumOfProducts	Integer	Number of bank products
HasCrCard	Binary	Has credit card (1/0)
IsActiveMember	Binary	Active customer (1/0)
EstimatedSalary	Decimal	Salary of customer
Exited	Binary	Churn status (1 = Yes, 0 = No)

5. Tools & Technologies

- **Excel**: Data cleaning, transformation, pivot tables
- **Power BI**: Data modeling, DAX, dashboard creation

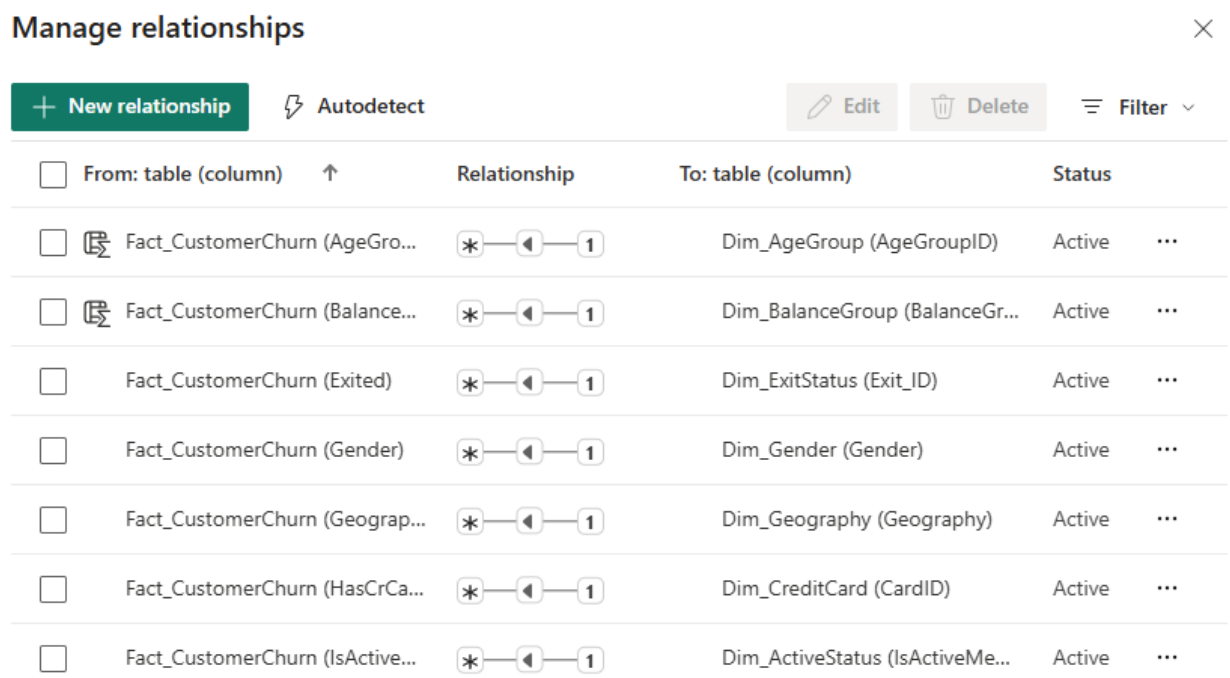
6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- Removed duplicates and null values
- Standardized column formats
- Created new columns:
    - Age Group
    - Balance Group
- Filtered irrelevant data
- Created Pivot Tables for initial analysis
- Converted dataset into structured format

7. Data Modelling and DAX (Power BI)

Data Model:



From: table (column)	Relationship	To: table (column)	Status
Fact_CustomerChurn (AgeGro...	* — 1	Dim_AgeGroup (AgeGroupID)	Active
Fact_CustomerChurn (Balance...	* — 1	Dim_BalanceGroup (BalanceGr...	Active
Fact_CustomerChurn (Exited)	* — 1	Dim_ExitStatus (Exit_ID)	Active
Fact_CustomerChurn (Gender)	* — 1	Dim_Gender (Gender)	Active
Fact_CustomerChurn (Geograp...	* — 1	Dim_Geography (Geography)	Active
Fact_CustomerChurn (HasCrCa...	* — 1	Dim_CreditCard (CardID)	Active
Fact_CustomerChurn (IsActive...	* — 1	Dim_ActiveStatus (IsActiveMe...	Active

- The model follows a **Star Schema** architecture, centering on the `Fact_CustomerChurn` table surrounded by multiple specialized dimension tables (e.g., `Dim_AgeGroup`, `Dim_Geography`).
- All identified relationships utilize a **Many-to-One (*:1)** cardinality, ensuring that each transaction in the fact table maps to a unique descriptive attribute in the dimensions.

DAX Measures:

DAX for creating Agegroup Column in Fact table

```
AgeGroup =
SWITCH(TRUE(),
   Fact_CustomerChurn[Age] <= 25, "18-25",
   Fact_CustomerChurn[Age] <= 35, "26-35",
   Fact_CustomerChurn[Age] <= 45, "36-45",
   Fact_CustomerChurn[Age] <= 60, "46-60",
   "60+"
)
```

Dax for creating AgeGroupID in Fact table

```
AgeGroupID =
SWITCH(TRUE(),
    Fact_CustomerChurn[Age] <= 25, 1,
    Fact_CustomerChurn[Age] <= 35, 2,
    Fact_CustomerChurn[Age] <= 45, 3,
    Fact_CustomerChurn[Age] <= 60, 4,
    5
)
```

DAX for creating Balance Group Column in Fact table

```
BalanceGroup =
SWITCH(TRUE(),
    Fact_CustomerChurn[Balance] = 0, "Zero",
    Fact_CustomerChurn[Balance] <= 50000, "Low",
    Fact_CustomerChurn[Balance] <= 100000, "Medium",
    "High"
)
```

DAX for creating Balance GroupID Column in Fact table

```
BalanceGroupID =
SWITCH(TRUE(),
    Fact_CustomerChurn[Balance] = 0, 1,
    Fact_CustomerChurn[Balance] <= 50000, 2,
    Fact_CustomerChurn[Balance] <= 100000, 3,
    4
)

Churn Rank =
RANKX(
    ALL(Dim_Geography[Geography]),
    [Churned Customers],
    ,
    DESC
)

Churn Contribution % =
DIVIDE(
    [Churned Customers],
    CALCULATE(
```

```
        [Churned Customers],
        ALL(Dim_Geography[Geography])
    )
)
```

DAX for creating Dim_AgeGroup Table

```
Dim_AgeGroup =
DATATABLE(
    "AgeGroupID", INTEGER,
    "AgeGroup", STRING,
    {
        {1, "18-25"},
        {2, "26-35"},
        {3, "36-45"},
        {4, "46-60"},
        {5, "60+"}
    }
)
```

AgeGroupID	AgeGroup
1	18-25
2	26-35
3	36-45
4	46-60
5	60+

DAX for creating Dim_BalanceGroup Table

```
Dim_BalanceGroup =
DATATABLE(
    "BalanceGroupID", INTEGER,
    "BalanceGroup", STRING,
    {
        {1, "Zero"},
        {2, "Low"},
        {3, "Medium"},
        {4, "High"}
    }
)
```

BalanceGroupID	BalanceGroup
1	Zero
2	Low
3	Medium
4	High

DAX for retrieving GeographyID from Dim_geography to Fact table

```
GeographyID =
LOOKUPVALUE(
    Dim_Geography[Geography_ID],      -- Column to return
    Dim_Geography[Geography],    -- Search column in dimension
    Fact_CustomerChurn[Geography]    -- Column in fact table to match
)
```

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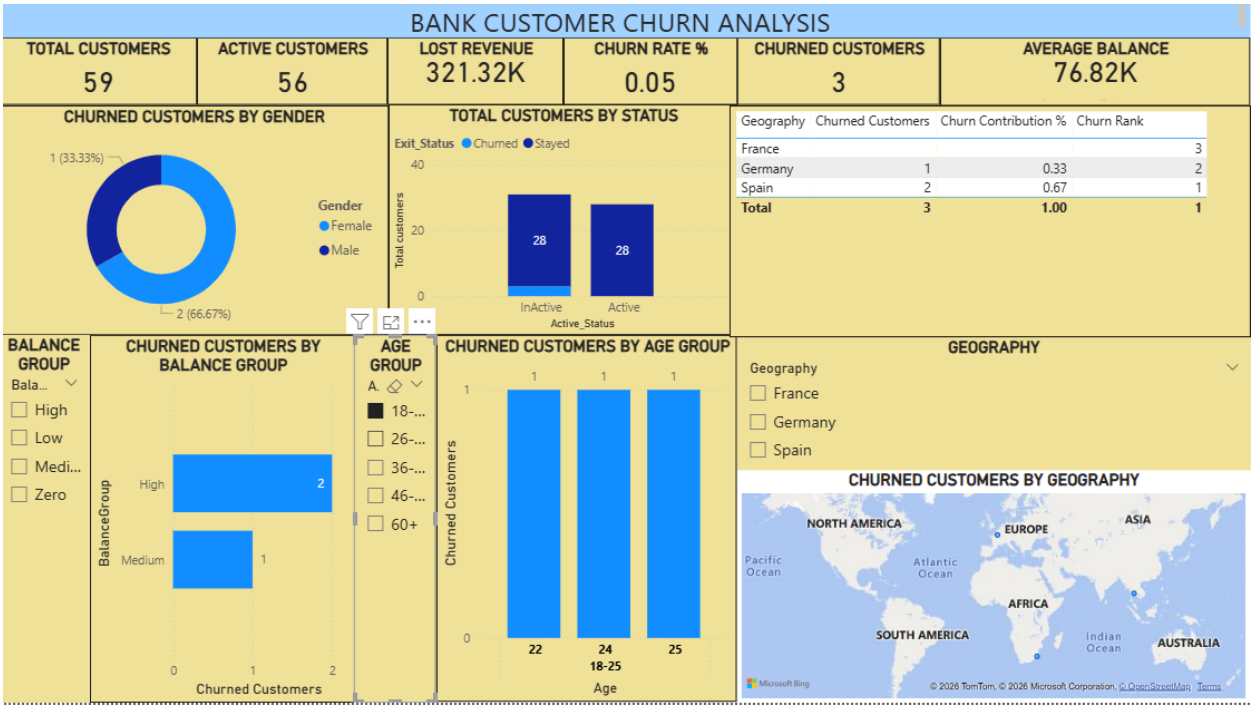
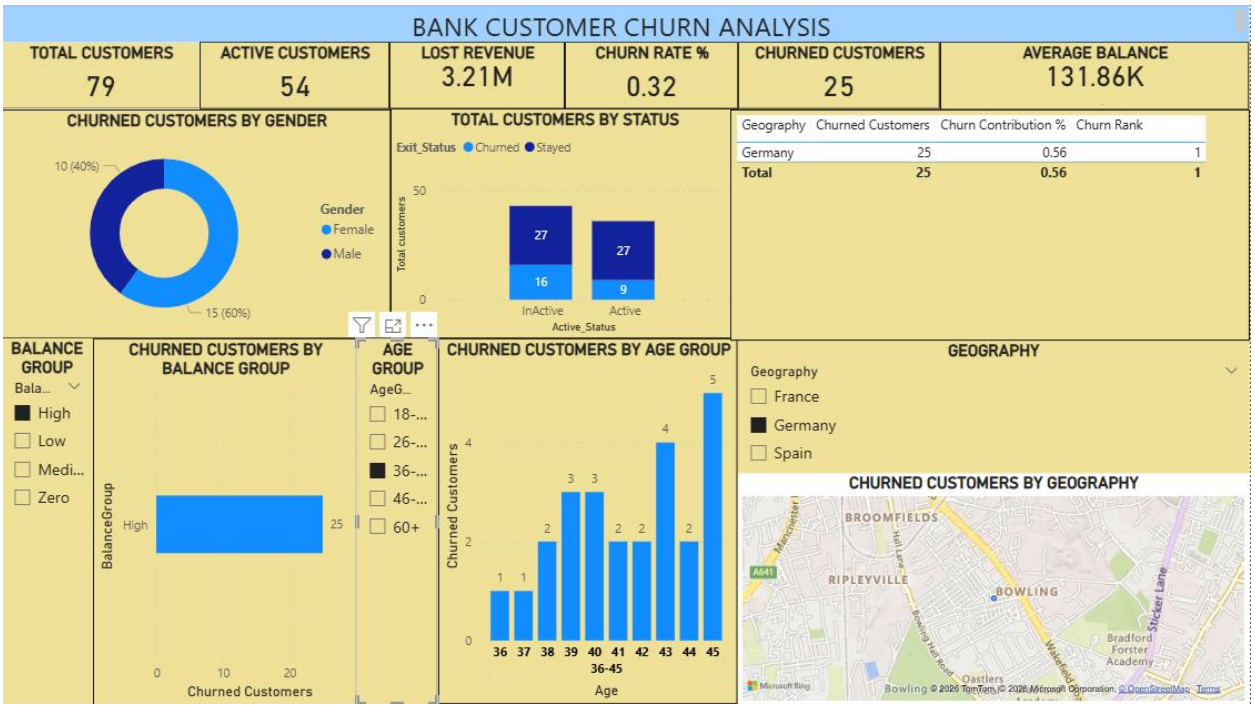
8. DashBoard (Power BI)





9. Analysis and Key findings:

Descriptive Analysis (What Happened)

- Calculated total customers and overall churn rate
- Analyzed churn distribution across geography (France, Germany, Spain)
- Compared churn rates between male and female customers
- Evaluated churn based on age groups and tenure
- Assessed account balance and product usage patterns
- Identified active vs inactive customer proportions

Diagnostic Analysis (Why It Happened)

- Identified higher churn among customers with low tenure
- Analyzed impact of low credit scores on churn probability
- Compared churn behavior between active and inactive members
- Examined effect of number of products on customer retention
- Investigated high churn in specific regions (e.g., Germany)
- Detected patterns where high balance customers still exited

Predictive Analysis (What Will Happen)

- Predicted likelihood of churn based on customer attributes
- Identified high-risk customer segments using behavior patterns
- Forecasted churn trends across different age and tenure groups
- Estimated future churn rate based on historical data
- Modeled impact of engagement (activity level) on churn
- Highlighted potential churn among low-product users

Prescriptive Analysis (What Should Be Done)

- Target high-risk customers with personalized retention offers
- Improve engagement strategies for inactive customers
- Offer incentives for customers with fewer products
- Focus retention campaigns on high-churn regions
- Enhance services for customers with low credit scores
- Develop loyalty programs to retain long-term customers

Key findings:

- Customers aged **30–50** show higher churn
- High balance customers are more likely to churn
- Germany has higher churn compared to other countries
- Inactive customers have higher churn rate

10. Conclusion

- The project provides a comprehensive analysis of customer churn using Power BI, combining descriptive, diagnostic, predictive, and prescriptive approaches.
- It identifies key factors influencing churn such as credit score, tenure, activity status, and number of products.
- The analysis highlights higher churn rates among inactive customers and those with lower engagement levels.
- Geographic trends reveal variations in churn behavior across different regions.
- Customer demographics like age and gender also contribute to churn patterns.
- Predictive insights help identify high-risk customers likely to leave the bank.
- The project enables proactive decision-making through data-driven forecasting.
- Prescriptive strategies such as targeted offers and engagement plans improve customer retention.
- Interactive dashboards enhance monitoring of KPIs and business performance.
- Overall, the project establishes a scalable and effective framework to reduce churn and improve customer satisfaction.